

PROBLEM SET 7

This is a practice problem set. You do not need to turn it in.

1. Consider the following game between a plaintiff (player 1) and a defendant (player 2) in a civil suit. The plaintiff knows the evidence in her hands; more precisely, she knows whether or not she will win the case if it goes to trial. Let θ_w represent the plaintiff's type that wins the trial and θ_L the type that loses the trial. The defendant does not have this information, but knows that the plaintiff knows her type (that is, who wins the trial). The defendant has prior belief, which is a common knowledge, that the probability that the plaintiff wins (i.e. the plaintiff is of high type) is $1/3$. If the case goes to trial and the plaintiff wins, her payoff is 3, while the defendant's payoff is -4 . If the case goes to trial and the defendant wins, his payoff is 0 and the plaintiff's payoff is -1 . The plaintiff has only two possible actions: she can ask for either a low settlement of $m = 1$ or a high settlement $m = 2$ (she cannot refuse to offer a settlement and she cannot decide to dismiss the case). After an offer has been made, the defendant decides whether to accept or reject it. If the settlement offer is accepted, the plaintiff's payoff is m and the defendant's payoff is $-m$. If the offer is rejected, the case goes to trial.
 - (a) Draw the game tree depicting the extensive form of this game.
 - (b) Define a pure-strategy PBE for this game (that is which requirements a PBE needs to satisfy).
 - (c) List all the pure-strategy profiles. For each such profile, specify the posterior beliefs of the defendant as a function of m and check whether the combination of these beliefs and the profile is PBE. How many PBE do you find?
2. Consider a Spence signalling model in which there are 3 potential types of workers $\theta \in \{1, 2, 3\}$ where θ denotes the productivity of the worker. Suppose that the utility of type θ is

$$u(w, e, \theta) = w - \frac{e}{\theta}$$

where w is wages and e is education. Suppose that each type of worker is equally likely. Let $\mu(\theta|e)$ denote the firms' belief that the worker is of type θ conditional of observing education level e .

- (a) Find a least-cost fully separating PBE $(e^*(\theta), w^*(e), \mu^*(\theta|e))$.
 - (b) Find a least-cost partially separating equilibrium $(e^{**}(\theta), w^{**}(e), \mu^{**}(\theta|e))$, in which $\theta = 1, 2$ pool and $\theta = 3$ separates.
 - (c) Find a least cost fully pooling equilibrium $(e^p(\theta), w^p(e), \mu^p(\theta|e))$.
 - (d) Pareto rank the equilibria from a), b) and c).
3. Econ Qualifier Summer 2011- Part III, Question 2